

K-Filter Ensemble GNN is All You Need: Extending Triple Filter Ensembles for Enhanced Spectral Graph Neural Networks

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Abstract

K-Filter Ensemble GNN (KFE-GNN) extends the *Triple Filter Ensemble* (TFE) architecture and introduces a bank of band-pass ensembles, each tuned to a specific frequency band. This design enables the model to capture low-, mid-, and high-frequency features directly, offering generalized performance on homophilic, heterophilic, and mixed structural patterns. Notably, we achieved 5% higher accuracy on the Squirrel dataset, a mixed dataset, while maintaining performance on graphs of differing structures.

1 Introduction

Spectral Graph Neural Networks (Spectral GNNs) are a subset of GNNs which operate in the frequency domain of the graph Laplacian and can reliably distinguish between homophilic and heterophilic structures. Recent work on the TFE-GNN (Duan et al., 2024) combines one low-pass and one high-pass branch to cover these cases.

However, many real graphs also contain important *mid-frequency* features and the previously proposed two-branch design can only approximate these signals indirectly. To address this gap, we extend the TFE idea to KFE: we add a set of band-pass kernels, each focused on a different frequency band. This design lets the model learn low-, mid-, and high-frequency patterns at the same time and fuse them with simple, learnable weights.

Motivation. In a simple line-graph study, we built a node signal $f = (L + M + H)$ that mixes low-, mid-, and high-frequency information. Repeated high-pass applications cleanly reveal the high-frequency H , and repeated low-pass applications isolate the low-frequency L . However, the mid-band M remains entangled with L , and a single polynomial low-pass ensemble can extract either M or L depending on its coefficients, but not both simultaneously. This limitation motivates a

KFE with dedicated band-pass branches or additional ensembles with a separation constraint, enabling simultaneous capture of each component and a more faithful decomposition before fusion (see [Appendix A](#) for details and figures).

2 Background

2.1 Spectral Graph Neural Networks

Spectral GNNs define graph convolution in the frequency domain of a graph Laplacian. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with adjacency $\mathbf{A}$, degree $\mathbf{D}$ and normalized Laplacian $\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$. With eigendecomposition $\mathbf{L}_{\text{sym}} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top$, a spectral filter g acts on features $\mathbf{X}$ as

$$g(\mathbf{L}_{\text{sym}}) \mathbf{X} = \mathbf{U} g(\mathbf{\Lambda}) \mathbf{U}^\top \mathbf{X},$$

which scales each spectral component of $\mathbf{X}$. Early methods such as ChebNet (Defferrard et al., 2016) approximate g with Chebyshev polynomials and later models like GPR-GNN (Chien et al., 2021) and BernNet (He et al., 2021) use other polynomial bases with learnable weights.

2.2 Feature Frequency

Feature frequency describes how fast node features vary over the graph. The spectral coefficients are $\hat{\mathbf{X}} = \mathbf{U}^\top \mathbf{X}$.

- **Low frequency:** smooth, large-scale patterns over the features of $\mathcal{V}$ across $\mathcal{E}$.
- **Mid frequency:** patterns that are neither very smooth nor highly oscillatory.
- **High frequency:** highly oscillatory over the features of $\mathcal{V}$ across $\mathcal{E}$.

2.3 Graph Filters

A *graph filter* is a linear operator designed to emphasize certain graph frequencies while attenuating

others. In this work, we focus on the following basic filters:

$$\begin{aligned}\mathbf{H}_{\text{lp}} &= \mathbf{L}_{\text{sym}}, & (\text{low-pass}) \\ \mathbf{H}_{\text{hp}} &= \mathbf{I} - \mathbf{L}_{\text{sym}}, & (\text{high-pass}) \\ \mathbf{H}_{\text{bp}} &= \mathbf{H}_{\text{hp}}^\alpha \mathbf{H}_{\text{lp}}^\beta, & (\text{band-pass})\end{aligned}$$

With $\alpha, \beta \in \mathbb{N}$ controlling the bandwidth of the band-pass filter. By raising these operators to powers in $\mathbb{N}$ and mixing them to create polynomials, we obtain a family of higher-order filters that provide increasing selectivity.

2.4 Triple Filter Ensemble GNN

TFE-GNN (Duan et al., 2024) addresses the limitations of a single polynomial filter by *ensembling* low-pass and high-pass branches and subsequently fusing them. Specifically, define

$$\text{TFE}_1 = \text{EM}_1\{\omega_0 \mathbf{I}, \dots, \omega_{K_{\text{lp}}} \mathbf{H}_{\text{lp}}^{K_{\text{lp}}}\}, \quad (1)$$

$$\text{TFE}_2 = \text{EM}_2\{\omega'_0 \mathbf{I}, \dots, \omega'_{K_{\text{hp}}} \mathbf{H}_{\text{hp}}^{K_{\text{hp}}}\}, \quad (2)$$

$$\text{TFE}_3 = \text{EM}_3\{\vartheta_1 \text{TFE}_1, \vartheta_2 \text{TFE}_2\}, \quad (3)$$

where $K_{\text{lp}}, K_{\text{hp}} \in \mathbb{N}$ and each EM_i denotes an ensemble combiner (a weighted sum or a gating mechanism). Given input features $\mathbf{X}$, TFE produces $\mathbf{Z} = \text{TFE}_3 \mathbf{X}$, which is then mapped to predictions $\hat{\mathbf{Y}}$ by a head network (an MLP with softmax).

This design accommodates both homophilic and heterophilic structures, while mid-frequency information is modeled only indirectly through the fusion of the two branches.

2.5 Homophily metric

We quantify structural homophily with the *edge homophily ratio* (EHR), the fraction of edges whose endpoints share a label:

$$\text{ehr} = \frac{|\{(u, v) \in E : Y_u = Y_v\}|}{|E|} \in [0, 1].$$

Higher values indicate stronger homophily while lower values indicate heterophily.

2.6 Main Datasets

According to (Ma et al., 2021), **Cora** is strongly homophilic ($\text{ehr} \approx 0.81$), while **Chameleon** (≈ 0.23), **Squirrel** (≈ 0.22), **Cornell** (≈ 0.30), and **Wisconsin** (≈ 0.21) are considered heterophilic.

Unlike Cora, which is dominated by homophily (low-frequency signals), Squirrel and Chameleon

are larger graphs with heterophilic local structures. Thus, their global edge homophily is low ($\text{ehr} \approx 0.22\text{--}0.23$), even though within neighborhoods one can find both homophilic and heterophilic patterns, which make them perfect for our evaluation.

3 Methodology

3.1 K-Filter Ensemble (KFE)

We extend the TFE idea to $K \geq 2$ learnable branches: LP, HP, and $K - 2$ mid-bands. Each branch filters features with a band-specific propagation operator and a learnable polynomial, and the band outputs are fused before a compact MLP.

Base filters. Using the symmetric η -normalized adjacency,

$$\mathbf{P}_{\text{sym}}^{(\eta)} \triangleq \mathbf{D}^{-\eta} \mathbf{A} \mathbf{D}^{-\eta},$$

we define the low- and high-pass filters as

$$\mathbf{H}_{\text{lp}} \triangleq \mathbf{P}_{\text{sym}}^{(0.5)}, \quad (4)$$

$$\mathbf{H}_{\text{hp}} \triangleq \mathbf{I} - \mathbf{P}_{\text{sym}}^{(\eta)}, \quad (5)$$

Mid-bands. We expose intermediate frequencies with

$$\mathbf{B}_b \triangleq \mathbf{H}_{\text{hp}}^{\alpha_b} \mathbf{H}_{\text{lp}}^{\beta_b}, \quad (6)$$

$$\mathcal{A} \triangleq [\mathbf{H}_{\text{lp}}, \mathbf{B}_1, \dots, \mathbf{B}_{K-1}, \mathbf{H}_{\text{hp}}]. \quad (7)$$

Per-band propagation. For band j with operator $\mathbf{A}_j$ and order K_j ,

$$\mathbf{Z}_j = \sum_{k=0}^{K_j} \omega_k^{(j)} \mathbf{A}_j^k \mathbf{X}, \quad (8)$$

with a learned coefficient vector $\{\omega_k^{(j)}\}_{k=0}^{K_j}$ per band.

Fusion and prediction. We define 2 combination methods for fusing the per-band filtered features:

$$\mathbf{Z}^{\text{sum}} \triangleq \sum_j \vartheta_j \mathbf{Z}_j, \quad (9)$$

$$\mathbf{Z}^{\text{con}} \triangleq \parallel_j (\vartheta_j \mathbf{Z}_j) \quad (10)$$

with a learned coefficient ϑ_j per band.

The final prediction head is

$$\mathbf{Z}_o \triangleq \text{MLP}(\mathbf{Z}). \quad (11)$$

Concatenation only increases the input width of the *first* linear layer in the MLP; subsequent layers keep their usual sizes.

3.2 Bandwidth–Decorrelation Penalty

Define the ℓ_2 -normalized embeddings

$$\hat{\mathbf{Z}}_b \triangleq \frac{\mathbf{Z}_b}{\|\mathbf{Z}_b\|_2}$$

where $b \in \{1, \dots, K\}$. We add a cosine-similarity penalty that discourages redundancy among the bands. With

$$\mathcal{S} \triangleq \{(b, b') : b, b' \in [K], b < b'\}$$

the penalty is

$$\mathcal{L}_{\text{pen}} = \sum_{(b, b') \in \mathcal{S}} \langle \hat{\mathbf{Z}}_b, \hat{\mathbf{Z}}_{b'} \rangle.$$

The gradients from this loss are propagated only to the midbands, allowing the HP and LP branches to operate as in the previous architecture. The total training objective augments the supervised loss with this penalty:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mu \mathcal{L}_{\text{WD}} + \lambda \mathcal{L}_{\text{pen}}.$$

where $\mathcal{L}_{\text{CE}}$ is the cross entropy loss and $\mathcal{L}_{\text{WD}}$ is the standard weight decay.

4 Results

To evaluate our model, we devised 3 experiments, each measuring the effect of different hyperparameters on the performance of the model. Each configuration was evaluated with 20 monte-carlo iterations.

4.1 Experiment 1 - Changing λ

For this experiment, we chose to focus on the Squirrel dataset which incorporates a mixture of high and low frequency information. Table 1 showcases results over varying values of λ and 1-2 additional mid-bands (3-4 total) with varying bandwidths. All other hyper-parameters can be found in Appendix B, Table 4.

We see better results than the TFE model, which averaged 72.27% accuracy, across the board, indicating that the addition of mid bands can indeed improve classification accuracy.

Table 1: Experiment 1 Results

(α, β) Lambda	1, 3	1, 5	1, 3; 3, 1	1, 3; 3, 5	1, 5; 3, 5
0	73.29	74.25	72.91	72.62	72.52
0.001	76.17	73.67	73.48	70.98	73.00
0.5	75.79	73.67	74.54	73.67	75.79
1	74.92	74.25	74.15	74.35	73.67
2	74.06	74.06	75.60	72.91	73.87
3	75.4	74.92	73.87	73.67	74.54
4	75.12	77.23	74.06	76.84	75.79
5	74.54	75.5	74.92	74.44	73.58
10	75.5	75.21	73.48	71.56	73.87

4.2 Experiment 2 - Polynomial Degrees

In this experiment we focus on the Chameleon dataset which has similar properties to Squirrel. We run the evaluations with two sets of polynomial degrees. In Table 2a, all polynomials are generically assigned a degree of 5. In Table 2b we optimize the degrees according to the **ehr** of Chameleon such that we obtain a degree of 6 for the low-pass polynomial, a degree of 7 for all mid-band polynomials and a degree of 8 for the high-pass polynomial. All other hyper-parameters can be found in Appendix C, Table 5.

We see that careful tuning of the polynomial degrees has some effect on model performance such that overall, the results in Table 2b are slightly better. Additionally, we see a slight increase in performance when compared to TFE-GNN, 77.89% as opposed to 77.16%.

Table 2: Experiment 2 Results

(a) Generic Degrees				
Lambda	3 Bands		4 Bands	
	1,3	1,5	1,3;3,1	1,3;3,5
0	75.05	75.49	74.84	75.05
0.001	75.05	74.84	74.83	75.05
0.5	75.71	76.37	75.93	76.15
1	75.27	75.93	75.71	75.27
5	73.96	73.52	72.87	73.30
10	71.77	72.65	64.11	70.67
(b) Tuned Degrees				
Lambda	3 Bands		4 Bands	
	1,3	1,5	1,3;3,1	1,3;3,5
0	74.61	77.02	73.74	76.58
0.001	74.61	74.17	74.39	76.36
0.5	76.58	77.89	74.39	76.80
1	74.61	76.80	75.49	74.61
5	77.46	76.80	77.24	75.27
10	74.17	76.36	76.14	77.02

Table 3: Experiment 3 Results

(a) $\eta = 0.3$

Dataset	3 Bands						4 Bands					
	1,3			1,5			1,3;3,1			1,3;3,5		
	0.5	λ 1	10	0.5	λ 1	10	0.5	λ 1	10	0.5	λ 1	10
Wisconsin	56.25	91.25	86.25	28.75	92.5	90	98.75	100	31.25	97.5	100	28.75
Cornell	70.49	80.33	80.33	55.74	73.77	65.57	36.07	55.74	65.57	81.97	57.38	44.26
Chameleon	56.46	65.86	65.86	42.67	55.58	55.82	68.05	71.77	73.74	67.18	71.33	68.05
Cora	89.33	90.14	88.51	89	89.33	88.83	88.83	89.33	89.16	88.67	90.14	90.31

(b) $\eta = 0.5$

Dataset	3 Bands						4 Bands					
	1,3			1,5			1,3;3,1			1,3;3,5		
	0.5	λ 1	10	0.5	λ 1	10	0.5	λ 1	10	0.5	λ 1	10
Wisconsin	75	78.75	86.25	68.75	95	41.25	88.75	82.5	33.75	83.75	92.5	33.75
Cornell	81.96	77.04	85.24	80.32	83.60	75.40	81.96	78.68	90.16	80.32	85.24	42.62
Chameleon	56.46	66.96	66.96	42.67	67.83	55.82	68.05	72.65	74.61	67.17	69.58	72.64
Cora	89.82	88.51	89.66	88.67	89.82	89.33	89.66	89.49	88.17	89.66	89.98	88.34

4.3 Experiment 3 - Changing η

In this experiment, we investigate how η influences classification accuracy on four datasets: Wisconsin, Cornell, Chameleon, and Cora. We test two values of η , 0.3 and 0.5, under various band configurations and regularization strengths. All other hyper-parameters can be found in [Appendix D](#), Table 6. The results in Tables 3a and 3b confirm that η plays a key role in adapting spectral filters to different graph structures, and must be adjusted for each dataset to achieve the best accuracy.

5 Discussion and Conclusions

Across strongly homophilic datasets (e.g., Cora) and strongly heterophilic ones (e.g., Cornell, Wisconsin), KFE-GNN achieves performance comparable to the baseline TFE-GNN. This demonstrates that introducing additional band-pass filters does not degrade accuracy when graphs are dominated by either low- or high-frequency signals. The stability of results in these regimes, highlighted in section 4.3, show that the architecture generalizes well and maintains robustness under extremes of homophily or heterophily.

The clearest benefit of our approach emerges on the Squirrel dataset, which combines both homophilic and heterophilic relations. In this setting, described in section 4.1, KFE-GNN reaches **77.23%** accuracy with three bands, substantially surpassing the baseline TFE result of 72.27%. This

gain aligns with our core hypothesis: by explicitly modeling mid-band components, the network captures structural information that cannot be represented by pure low- or high-pass filters. The improvement is relatively consistent across nearby configurations, underscoring that the design is not only effective but also somewhat robust to hyper-parameter choice when compared to TFE-GNN.

These findings validate the central motivation of KFE-GNN: incorporating mid-frequency filters enables the model to handle graphs where structural patterns are mixed, which are common in real-world networks. On purely homophilic or heterophilic datasets, the method remains competitive, while on mixed datasets it provides measurable improvements. We conclude that KFE-GNN extends the flexibility of spectral GNNs by bridging the gap between low- and high-frequency analysis, offering a promising direction for future spectral designs that adapt to diverse graph structures.

6 Limitations and Future Work

Similarly to the original TFE model, this model is sensitive to the choice of hyperparameters, especially the building block filters (values for η , α and β). As we have demonstrated, in our current state, the model requires a certain amount of experience in choosing these values as well as the polynomial degrees. Using the original paper and the supplied hyperparameters, we were nonethe-

less able to achieve some positive results, however this is still a major limitation when considering the model for widespread use.

Additionally, without the original paper as a guideline, our analysis would have relied mainly on the standard homophily metric (**ehr**), which we found to be too coarse to capture the full spectrum of structural signals in a graph. Datasets such as Chameleon and Squirrel share nearly identical **ehr** values but differ significantly in their mid-frequency behavior. This explains why mid-band filters are sometimes effective and sometimes not, highlighted in the difference in performance gain between section 4.1 and section 4.2 when compared to TFE-GNN ($\sim 5\%$ and $\sim 1\%$ respectively).

We believe a promising solution is to design richer spectral metrics that move beyond edge-level homophily and explicitly quantify mid-band structure, e.g. by measuring agreement between endpoints of simple paths at different lengths. Such descriptors could provide more reliable guidance for spectral GNN design which would allow us to have a better understanding of spectral GNNs and improve them.

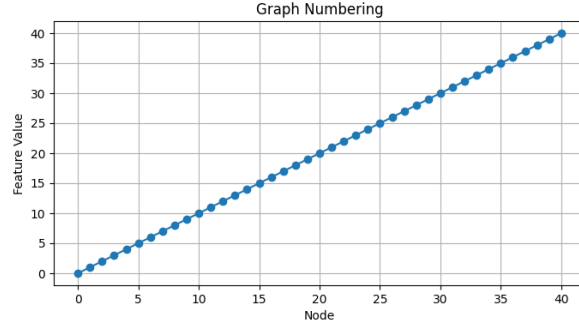
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Appendix A. Motivation

We present a toy example to motivate our extension of the TFE architecture. For clear visualization, we refer to the case of a line graph, which allows us to consider analogous concepts from one-dimensional signal processing.

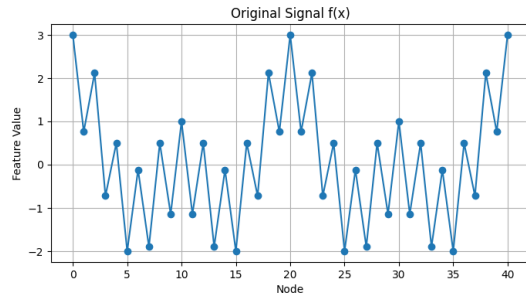
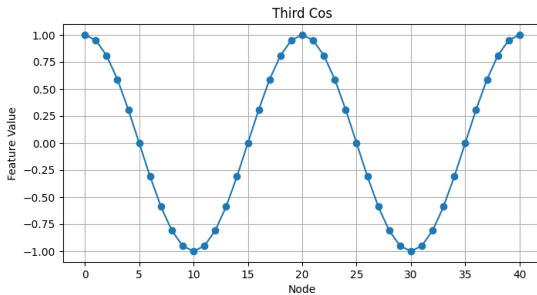
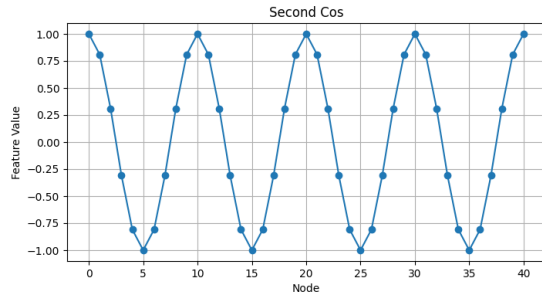
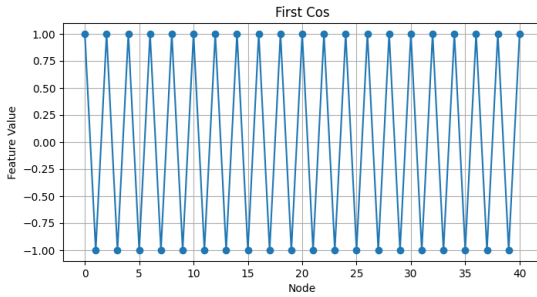
We define a line graph of length 41 and one-dimensional features. we number each node such that we create an artificial ordering on the unordered graph and get "initial features" which resemble a linearly increasing function:



We define transforms on the numbering:

$$c_1(x) = \cos(\pi x), c_2(x) = \cos(\frac{\pi x}{5}), c_3(x) = \cos(\frac{\pi x}{10})$$
$$f(x) = c_1(x) + c_2(x) + c_3(x)$$

Such that we get the following initial features:

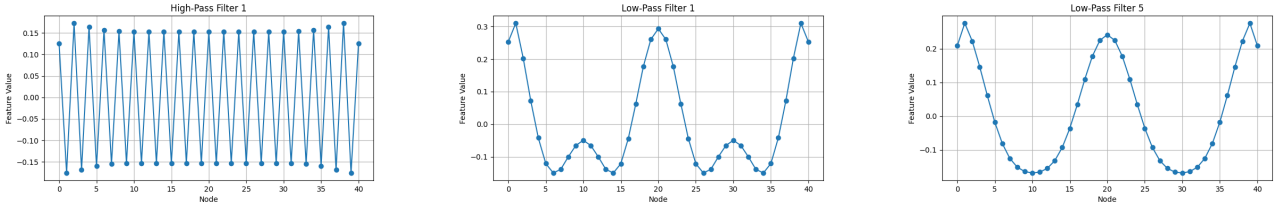


We have created a graph signal f clearly comprised of high frequency (c_1), mid frequency (c_2) and low frequency (c_3) components, which we separate using LP , BP and HP filters. A possible label which would require this separation could be $Y(v) = \underset{i}{argmax}\{c_i(v)\}$, where v represents the initial node numbering.

We use the HP and LP building block filters defined in the original paper:

$$\begin{aligned} HP &= L_{sym} \\ LP &= I - L_{sym} \end{aligned}$$

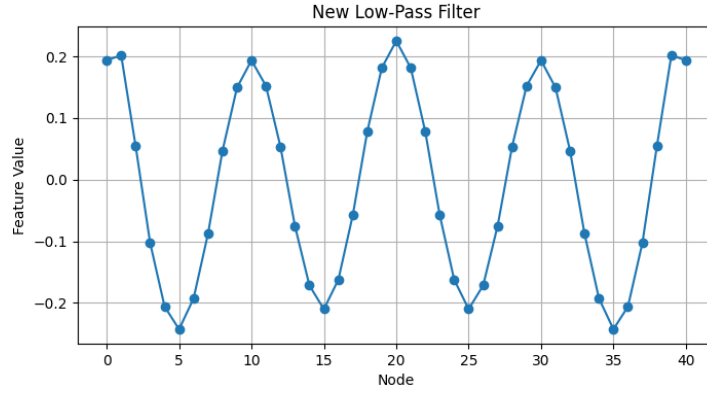
Where L_{sym} is the normalized symmetric graph laplacian. We apply each filter numerous times and show intermediate results:



We see a clear separation of c_1 and c_3 from f , however, we are unable to completely separate c_2 from c_3 .

In order to separate c_2 , we present a primitive form of midband filtering, using a mixture of the LP blocks similarly to the effect achieved by the third ensemble in the original paper:

$$BP = 5(I - L_{sym})^{(20)} - 4(I - L_{sym})^{(100)} = 5LP^{(20)} - 4LP^{(100)} = 5LP^{(20)}(I - 0.8LP^{(80)})$$



As expected, the previous architecture is able to separate all of c_1, c_2 and c_3 from f , however, very notably, it cannot separate all of them simultaneously. Since we used the LP polynomial to separate both c_2 and c_3 with different coefficients, when running the model, the output of the polynomial ensemble could be either c_2, c_3 or some mixture of the two, but not both. This provides the motivation for our model extension.

Appendix B. Hyperparameters for Squirrel Experiment

Table 4: Hyperparameter Settings for Exp. 1

Optimizer	Poly. Deg.	Epochs	MLP Layers	MLP Dim.	Drop _{pro}	Drop _{lin}	lr _{ada}	lr _{lin}	μ	η	Combine Method
Adam	5 (for all filters)	1000	2	512	0.6	0	0.1	0.005	0.05	0.3	sum

Appendix C. Hyperparameters for Chameleon Experiment

Table 5: Hyperparameter Settings for Exp. 2

Optimizer	Epochs	MLP Layers	MLP Dim.	Drop _{pro}	Drop _{lin}	lr _{ada}	lr _{lin}	μ	η	Combine Method
Adam	1000	2	512	0.4	0.3	0.1	0.005	0.05	0.3	sum

Appendix D. Hyperparameters for η Experiment

Table 6: Hyperparameter Settings for Exp. 3

Optimizer	Poly. Deg.	Epochs	MLP Layers	MLP Dim.	Drop _{pro}	Drop _{lin}	lr _{ada}	lr _{lin}	μ	Combine Method
Adam	5 (for all filters)	2000	3	1024	0.6	0	0.1	0.005	0.05	sum